# Phishing in the Digital Age: Surveying Public Awareness and Leveraging AI for Defense

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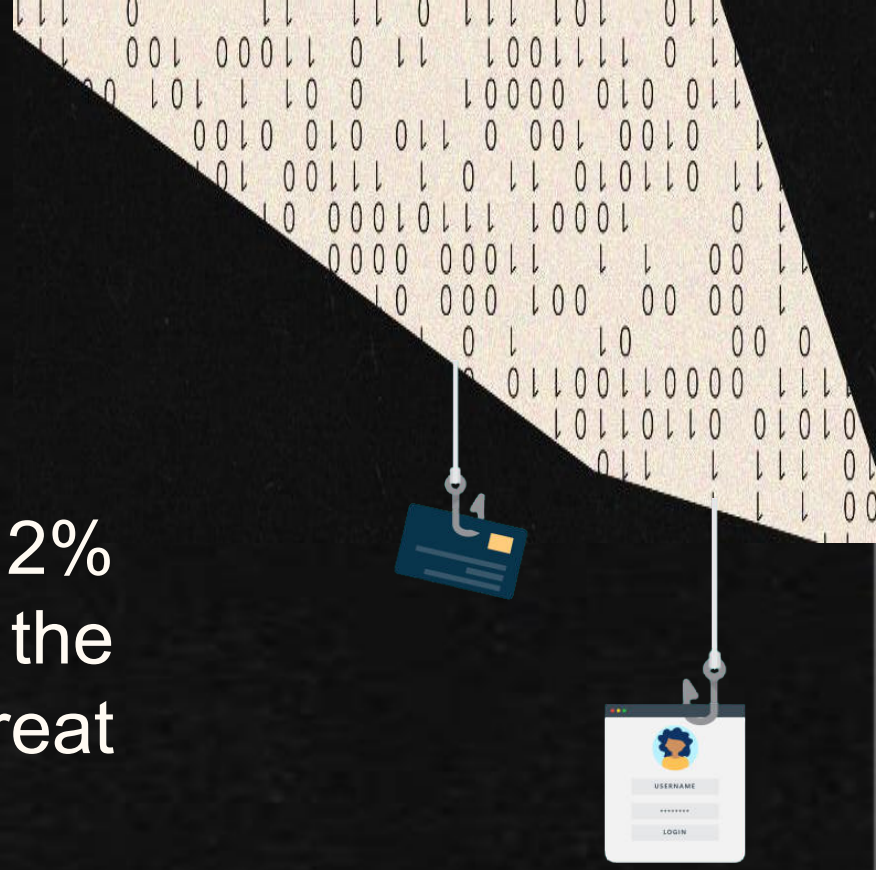
# *Background*

**Increase in attacks globally**

- 2023: phishing attacks surged by 58.2% compared to the previous year, reflecting the growing sophistication and reach of threat actors.

**Role of Public Awareness**

- 96% of working adults being aware of potential dangers, about 71% admitted to engaging in risky behaviors online.




Citation of stats

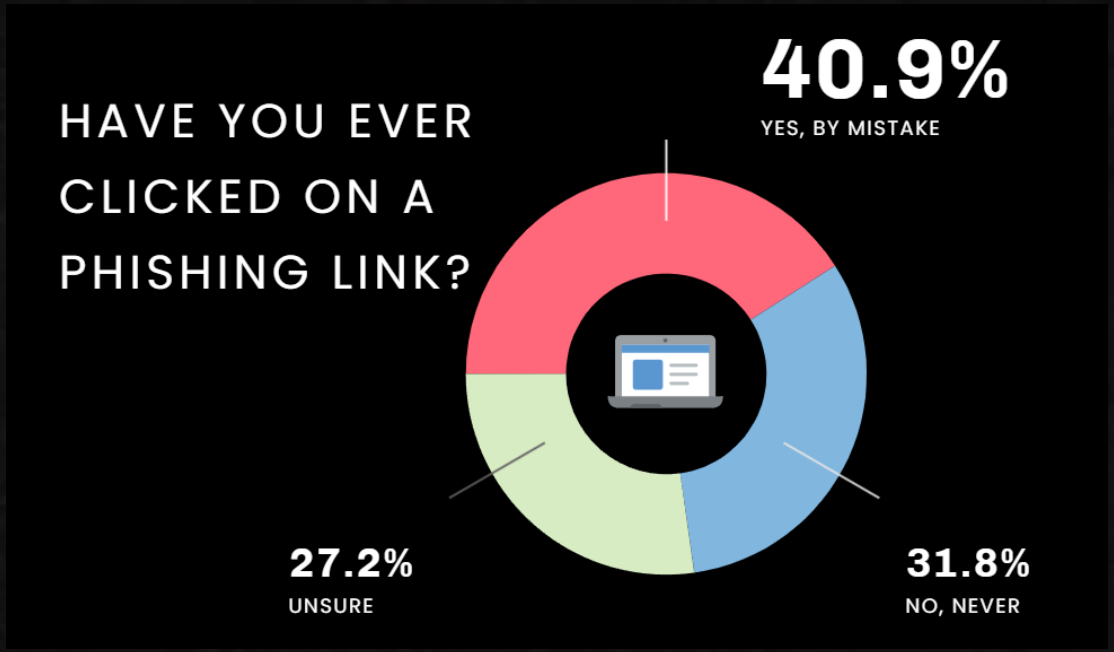
# Community Outreach and Survey

Survey Methodology

Outreach Activities

# Local Awareness and Survey

- Awareness Levels
- Common Threats Recognized
- Demographic Variations



# Mitigations

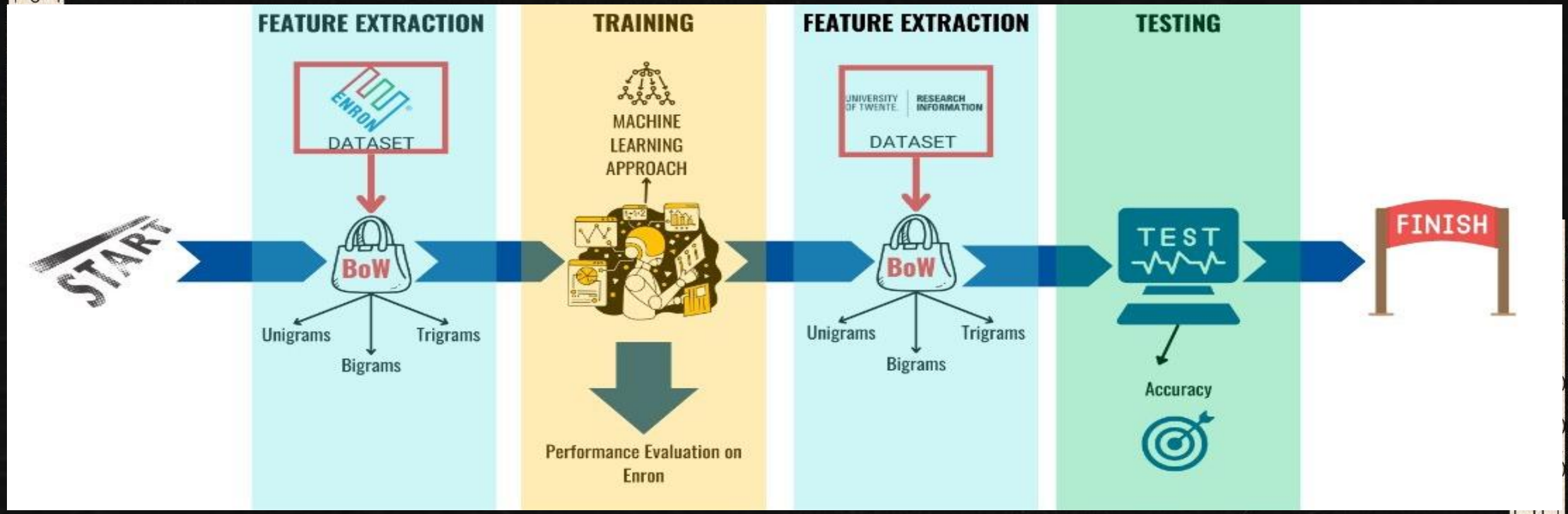
- AI-generated phishing countermeasures

- Mitigation Strategies
    - Education and Training
    - Implementing Multi-Factor Authentication
    - Combating with AI models

# Problem Statement

- How effective are AI phishing detection models trained on historical datasets in identifying modern, AI-driven phishing attacks with rapidly evolving tactics?

# Model Flowchart



# Feature Extraction

"Hello, we are happy to present at NCUR!"

Unigrams

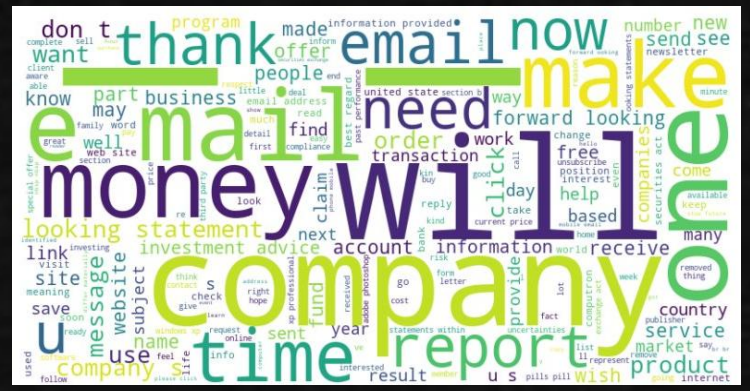
| hello | we | are | happy | to | present | at | ncur |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

Bigrams

| hello we | we are | are happy | happy to | to present | present at | at ncur |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |

Trigrams

| hello we are | we are happy | are happy to | happy to present | to present at | present at ncur |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |




Enron (2004)

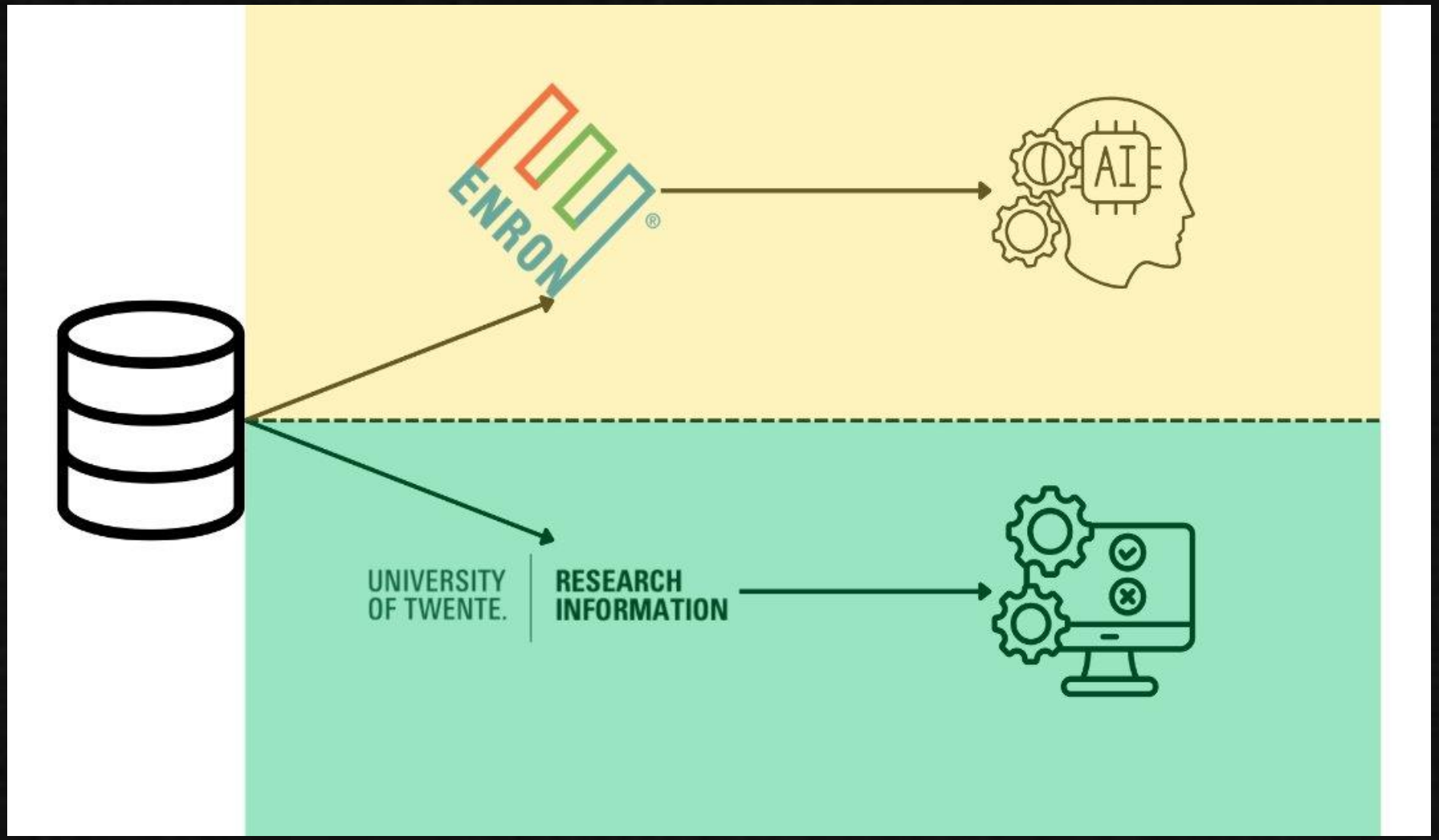
Phishing Validation (2024)

# Training Performances

- 80:20 split on Enron dataset

| Model | Accuracy |
|---|---|
| Naïve Bayes | 99.15% |
| Logistic Regression | 98.59% |
| Support Vector Machine | 95.89% |
| Random Forest | 98.10% |
| XGBoost | 97.39% |

# Testing Phase

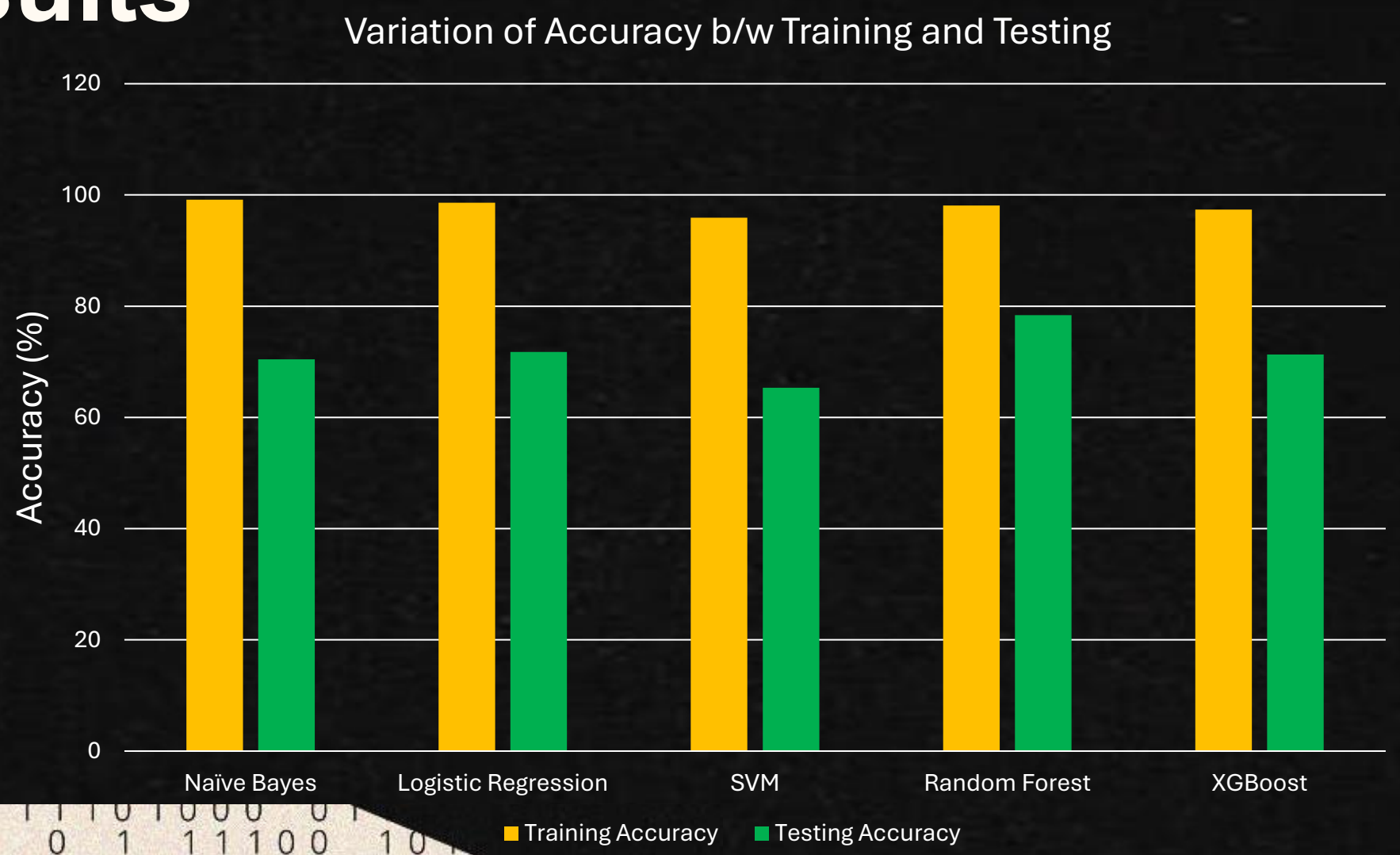


# Testing Performance

- Training on Enron dataset, testing on Phishing Validation dataset

| Model | Accuracy |
|---|---|
| Naïve Bayes | 70.45% |
| Logistic Regression | 71.75% |
| Support Vector Machine | 65.35% |
| Random Forest | 78.4% |
| XGBoost | 71.3% |

# Results


Variation of Accuracy b/w Training and Testing
Accuracy (%)
120
100
80
60
40
20
0
Naïve Bayes
Logistic Regression
SVM
Random Forest
XGBoost
Training Accuracy
Testing Accuracy

# Conclusion and Recommendations

- Classical machine learning and constant adaptation
- Up-to-date datasets
- Machine Unlearning